

Solution Requirements

SMARTBRIDGE / SKILL WALL

Date: 29 September 2026 | Team ID: SWTID-2026-6777

Project: ComicCraft - AI Comic Story Creator using Gemini Models

Define Functional and Non-Functional Requirements:

FR = what the system should do. NFR = how well the system should operate.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	No login is required in the current MVP. The comic creation interface is directly accessible to users.	Low
2	Authorization levels	Single creator/user role. Users can access all comic creation tasks; no admin role is required in the MVP.	Low
3	External interfaces	FastAPI REST endpoints connect the web UI with Google Gemini models, which generate the story, characters, scripts, and dialogue.	High
4	Transactions processing	User story ideas are validated, routed to the selected task, processed by the Gemini model, and returned as comic content.	High
5	Reporting	The system presents the plot outline, character profiles, panel-by-panel script, dialogue, and scene descriptions. No external export is required in the MVP.	Medium
6	Business rules, etc.	Empty inputs are rejected; the selected task determines the module; each comic is organized into numbered panels with dialogue and scene descriptions.	High
7	Compliance to laws or regulations	Protect the Gemini API key on the backend using environment configuration. Avoid unnecessary user data collection.	Medium
8	Other	Provide clear validation and error messages, maintain separate AI modules, and expose core comic creation functions through REST APIs.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Health endpoint should respond quickly; AI comic generation should complete within normal model/network limits.	Health check < 1 second; AI tasks complete without timeout under normal conditions.
2	Scalability	Keep FastAPI endpoints and AI modules separated so additional instances or model resources can be added.	API and AI processing components can scale independently.

Solution Requirements

SMARTBRIDGE / SKILL WALL | ComicCraft - AI Comic Story Creator using Gemini Models

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
3	Security & Data Privacy	Keep Gemini API key server-side and never expose secrets in HTML/JavaScript or source control.	No API secrets in frontend or GitHub; validate inputs.
4	Reliability & Availability	Return clear validation errors and handle model/API failures without crashing the application.	Invalid input gives 4xx; failures return a controlled error.
5	Usability & Accessibility	Provide a simple responsive HTML/CSS interface with clear task choices and readable comic output.	All five core tasks are reachable and results are clearly displayed.
6	Other	Use modular Python files and automated tests for core API/module behavior.	Core modules remain independently testable; tests run before submission.